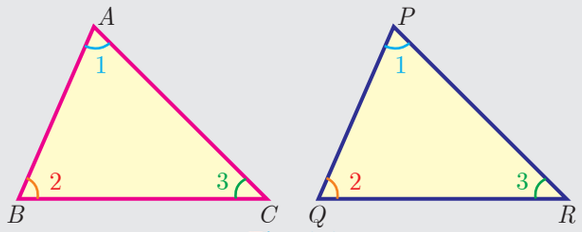
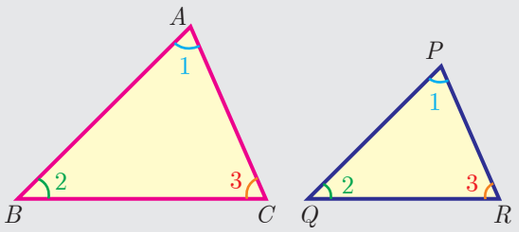
Congruency and similarity of triangles

Congruency is a particular case of similarity. In both the cases, three angles of one triangle are equal to the three corresponding angles of the other triangle. But in congruent triangles, the corresponding sides are equal. While in similar triangles, the corresponding sides are proportional.

Note

The traingles ABC and PQR are similar can be written as $\Delta ABC \sim \Delta PQR$

Congruent triangles	Similar triangles
 Fig. 4.4	 Fig. 4.5
$\Delta ABC \cong \Delta PQR$	$\Delta ABC \sim \Delta PQR$
$\angle A = \angle P, \angle B = \angle Q, \angle C = \angle R$.	$\angle A = \angle P, \angle B = \angle Q, \angle C = \angle R$
$AB = PQ, BC = QR, CA = RP$	$AB \neq PQ, BC \neq QR, CA \neq RP$
$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} = 1$	but $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} > 1$ *or* < 1
Same shape and same size.	Same shape but not same size.

Thinking Corner

1. Are square and a rhombus similar or congruent. Discuss.
2. Are a rectangle and a parallelogram similar. Discuss.

4.2.2 Criteria of Similarity

The following criteria are sufficient to prove that two triangles are similar.

AA Criterion of similarity

If two angles of one triangle are respectively equal to two angles of another triangle, then the two triangles are similar, because the third angle in both triangles must be equal. Therefore, AA similarity criterion is same as the AAA similarity criterion.

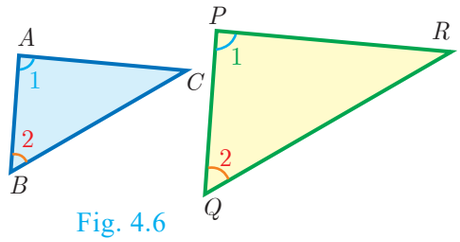


Fig. 4.6

So if $\angle A = \angle P = 1$ and $\angle B = \angle Q = 2$ then $\Delta ABC \sim \Delta PQR$.

SAS Criterion of similarity

If one angle of a triangle is equal to one angle of another triangle and if the sides including them are proportional then the two triangles are similar.

Thus if $\angle A = \angle P = 1$ and

$$\frac{AB}{PQ} = \frac{AC}{PR} \text{ then } \Delta ABC \sim \Delta PQR$$

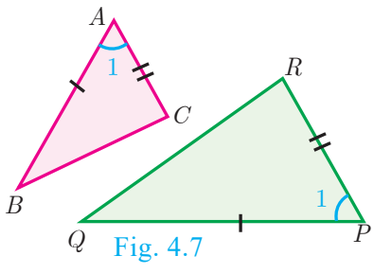


Fig. 4.7

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